

Year 8 Science – Booklet 15: Physics

Waves & Energy (Energy Transformation & Transfer) — ANSWERS

Note: despite sharing its title with Booklet 6, this booklet's actual content is entirely different – energy transformation, transfer and Sankey diagrams, not sound/waves. Answered on that basis.

Spot Check (p.27)

1. Potential energy.
2. Thermal (heat) energy.
3. Light energy can travel through a vacuum and doesn't need a medium; sound energy needs a medium (particles) to travel through and cannot travel through a vacuum.

Spot Check (p.29)

1. Thermal (heat) energy.
2. They vibrate more/faster, and this vibration is passed along to neighbouring particles by collision (conduction), transferring the heat through the metal.
3. Convection.

Energy transformation (workbook, p.120–121)

1. Thermal (heat) energy.
2. The Moon has no atmosphere (it's a vacuum), and sound needs a medium to travel through. With no air to carry sound waves, astronauts can't hear each other directly, so they use radios, which send radio waves that can travel through a vacuum.

3. *Sketching sound waves*

- a). Same pitch, much quieter: same number of humps (same frequency/wavelength) as the given wave, but with a much smaller amplitude (shorter peaks and shallower troughs).
- b). Just as loud, lower pitch: same peak height (amplitude) as the given wave, but fewer humps across the same width (a lower frequency, longer wavelength).

4. *Book on a high shelf*

- a). Gravitational potential energy.
- b). Kinetic energy.
- c). Sound energy and (thermal) heat energy.

5. *Odd one out*

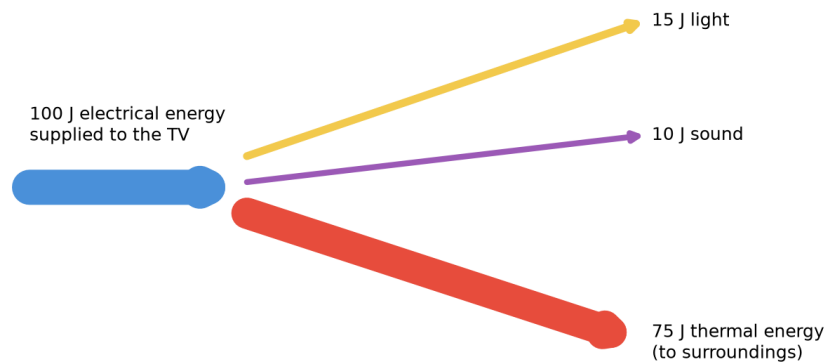
- a). **C – the string in a bow and arrow.** It stores elastic potential energy, whereas the other three (food, petrol, candle wax) all store chemical potential energy.
- b). Chemical potential energy.

Energy transfer (workbook, p.122–123)

1. *TV Sankey diagram (100 J in; 15 J light, 10 J sound given) (4 marks)*

Remaining energy = $100 - 15 - 10 = 75 \text{ J}$, transferred to the surroundings as thermal (heat) energy.

Sankey diagram: TV energy transfer



2. Saucepans and wooden spoons

- a). Metal (e.g. aluminium, stainless steel, or copper).
- b). Metals are good conductors of heat, so they let heat pass through easily and quickly from the hob to the food.
- c). Wood is a poor conductor of heat (an insulator), so it doesn't conduct heat up to the handle; a metal spoon would conduct heat along its length, making the handle too hot to hold safely.

3. Kettles, vacuums and cooling tea

- a). The heating element heats the water directly above it, making it less dense so it rises; cooler, denser water sinks to take its place and gets heated in turn – this convection current circulates and heats all the water efficiently.
- b). Radiation.
- c). By all three methods (3 marks): conduction (heat passes through the cup to anything touching it), convection (warm air above the tea rises, carrying heat away), and radiation (heat radiates directly from the hot tea into the surroundings, since it's hotter than them).

4. True or false

- a) Energy is not created or destroyed, just transferred – **True**
- b) Energy is sometimes used up, e.g. when a battery runs out – **False** (energy is transferred, not destroyed; the battery's store of chemical energy runs out, not energy overall)
- c) When energy is transferred it's not always in the form we want – **True**
- d) Most energy transferred by a light bulb goes into lighting the surroundings – **False** (most is wasted as heat, e.g. only 5 J of light out of 100 J for an old-style bulb)
- e) Energy is measured in joules, named after a person – **True** (James Prescott Joule)